

# Maddalena Torricelli

## Curriculum Vitae et Studiorum



### Contacts

maddalena.torricelli

### Digital Hub

maddalenatorricelli.me

LinkedIn

Google Scholar

Github

### Languages

IT (Native)

EN (Advanced)

ES, FR, DE (Basic)

### Code Stack

Python

Pandas, Numpy, Scipy, NetworkX, Gensim, Matplotlib, Seaborn, Pytorch

R

ggplot2, caret, glmnet

HTML, CSS, Javascript  
basic knowledge

LaTeX

DataViz

d3.js, Vega Lite, Google Data Studio

Intensive course on effective visualization of data with d3.js and Vega Lite (Big Dive, Torino (IT), 2020)

## Working Experience

2025 -

### Postdoctoral Research Associate

Northeastern University - Network Science Institute, USA

#### Sports analytics

Research topics: network science and data analytics applied to sports, with primary focus on football; leveraging complex systems methods for tactical analysis, team dynamics, and player performance.

Outreach and networking activities at major sports conferences: MIT Sloan Sports Analytics Conference (2025, 2026), MIT Sports Summit (2024, 2025, 2026), HUDL Performance Insights (2025), American Soccer Insights (ASI) Summit (Texas, 2026)

Hackathon organization in collaboration with Sports Analytics Club at Northeastern University and PySport

Visiting research period at ETH Zürich, in collaboration with Prof. Ulrik Brandes and the Social Networks Lab, focusing on a football analytics project

2021 - 2023

### Postdoctoral Research Associate

City University of London, UK

#### Climate change on social media

Research topics: polarization and misinformation spread on social media about climate change; quantifying social perception of climate change and natural disasters.

Teaching activity: Number Theory and Cryptography - Mathematics Dept.

Events organization: Data Natives (2022,2023), Complexity 72h (2023)

## Education

2017 - 2021

### PhD in Data Science and Computation

Alma Mater Studiorum di Bologna, Italy

Thesis: Embedded representations of social interactions

2015 - 2017

### MSc in Physics of Complex Systems

Università degli Studi di Torino, Italy

Thesis: Party cohesiveness in the Italian Parliament (1987-2013)

M.Sc.: 110/110 cum laude

## Selected publications

2026, May

### Evolution of spatial structure, passing network patterns, and gameplay intensity in elite women's and men's football (2020–2025)

Nature Scientific Reports

Paper resulting from Complexity 72h (2025).

Presented at ASI Summit, 31.01.2026, Houston,TX (USA)

2026, Jan

### The team support network: Quantifying off-ball impact in women's elite soccer through tracking data

SkillCorner Research Breakaways report

Presented at ASI Summit, 31.01.2026, Houston,TX (USA)

2024, May

### The role of interface design on prompt-mediated creativity in Generative AI

WEBSCI '24:

Proceedings of the 16th ACM Web Science Conference

Presented at WEBSCI Conference, 24.04.2024, Stuttgart (DE)

2023, Nov

### How does extreme weather impact the climate change discourse? Insights from the Twitter discussion on hurricanes

PLOS Climate

Presented at CCS Conference, 20.10.2023, Salvador, Bahia (BR)

2020, Apr

### weg2vec: Event embedding for temporal networks

Nature Scientific Reports

Presented at NetSci Conference, 29.05.2019, Burlington, VT (USA)

A full list of publications is available on my Google Scholar profile.

## Selected talks, projects and workshops

2026, Jan

### American Soccer Insights (ASI) Summit

Houston, TX (USA)

Two selected talks — see Publications section.

2025, Jun

### Complexity72h

Madrid (ES)

Participation as tutor at the Complexity72h workshop, mentoring students on a football analytics project.

2023, Jun

### Sònar+D

Barcelona (ES)

Collaboration with the EXPLAIN Project: analyzing lyrics and music of Sònar festival artists to detect patterns and track their evolution.